

Wireless Systems (3) - Medium Access Control protocols



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Figure-credits: some figures have been taken from slides published on the Web, by the following authors (in alphabetical order):

J.J. Garcia Luna Aceves (ucsc), James F. Kurose & Keith W. Ross, Jochen Schiller (fub), Nitin Vaidya (uiuc)

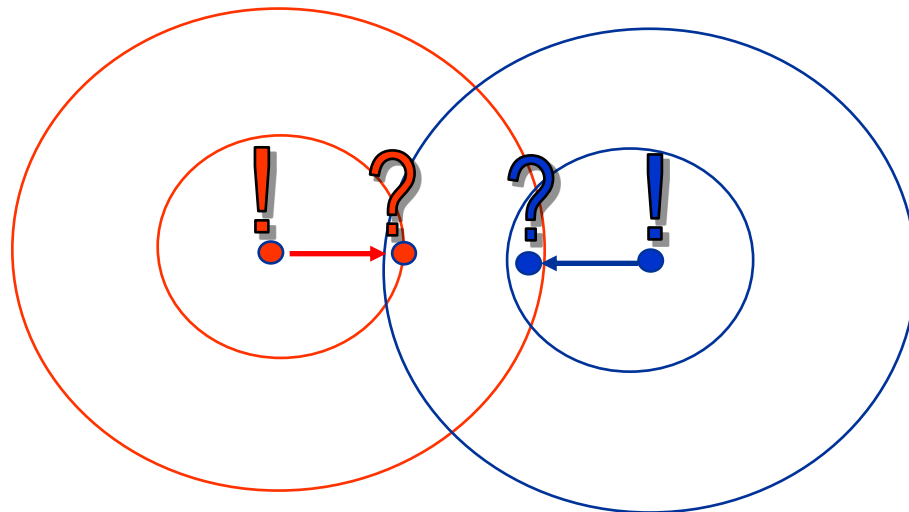
The MAC layer in wireless networks

- The **wireless** MAC layer roles
 - *Access control to shared channel(s)*
 - **Natural broadcast** of wireless transmission
 - Collision of signal: a time/**space** problem
 - Who transmits when? (**and where**)?
 - Avoid collisions (**no Collision Detection**)
 - *Scarce resources utilization*
 - **Channel capacity** and **battery power**
 - *performance and QoS*
 - System level and (**or vs?**) user level
 - *Frame organization, and intra-, inter-layer information management*
 - **Cross layering** principles for **adaptive behavior**?
 - Risk for “spaghetti design” [Kumar2003]

[Kumar2003] V. Kawadia, P.R. Kumar, "A Cautionary Perspective on Cross Layer Design", Submitted for publication, 2003
(<http://black1.csl.uiuc.edu/~prkumar/>)

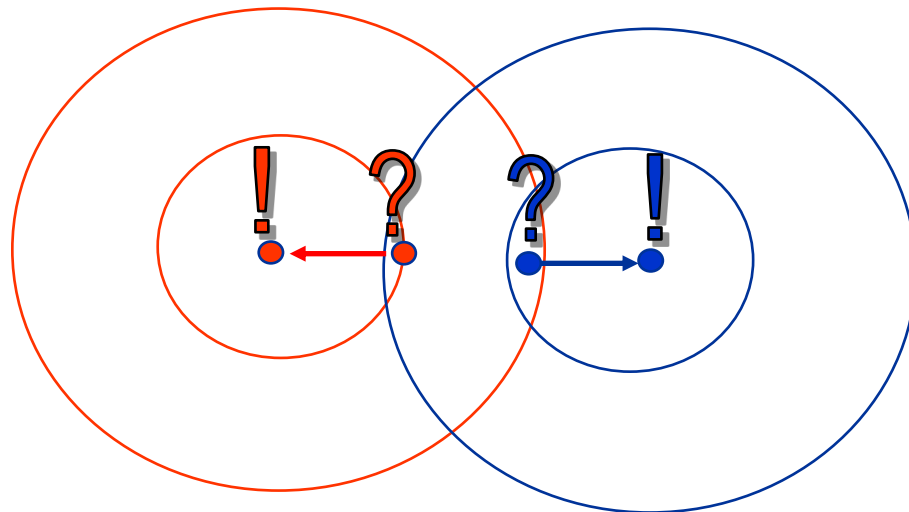
Collision of wireless signals

- **Collision** has **destructive effect on the receiver**
 - ...causes both channel and power waste
 - Collision detection is not practical in wireless systems
 - Collision avoidance/resolution + contention control on the sender
- Capture effect is possible
 - Exploited to enhance channel reuse, if possible
- Collision domain: set of nodes sharing the same channel
 - Space splitting, transitive relation

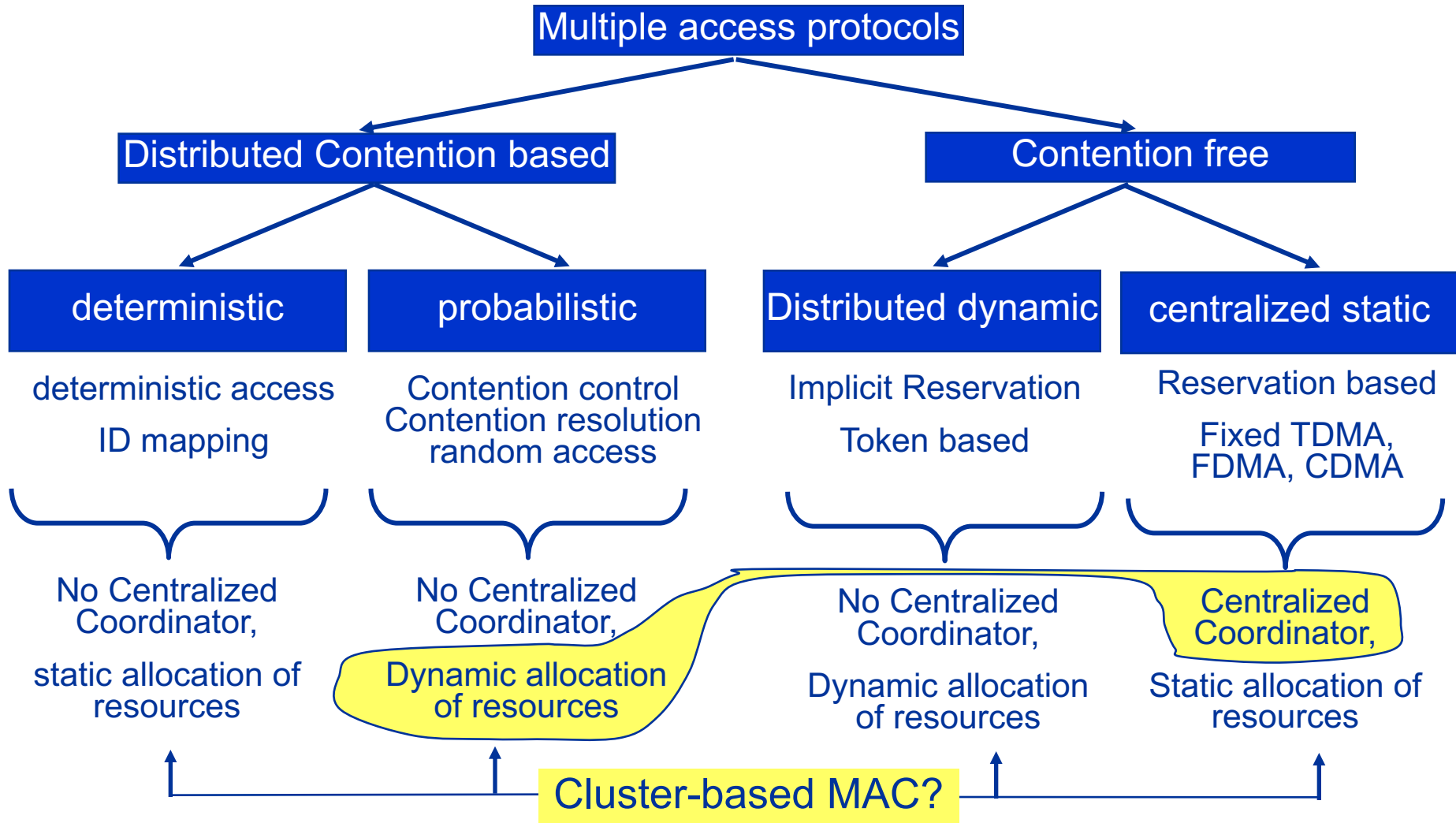


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Wireless MAC protocols' classification



Evolutionary perspective of distributed MAC

- Distributed, contention-based wireless MAC Problem:
 - the frame vulnerability (collision risk)
 - Needs resolution in distributed way (no centralized coordinator)
- let's analyze the **time domain first**
 - Aloha [Abramson1970]: no coordination
 - Slotted Aloha
 - CSMA [Kleinrock1975]: listen before to transmit
 - Slotted CSMA
 - CSMA/CD: listen before and while transmitting
 - (*unpractical in wireless scenarios*)
 - CSMA/CA + contention resolution (*reactive resolution of collisions*)
 - CSMA/CA + contention control (*preventive/reactive reduction of risk of collisions*)

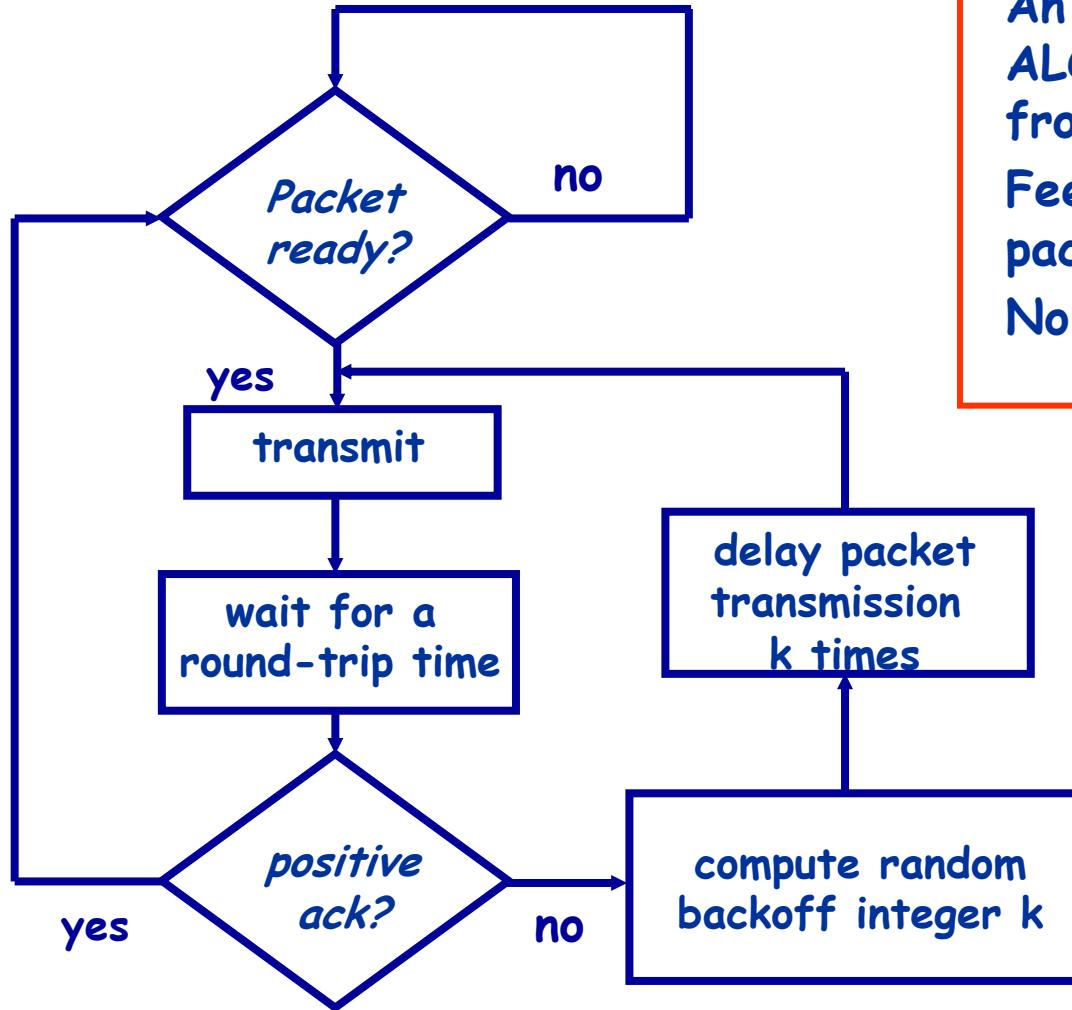
[Abramson1970]

N. Abramson, "The ALOHA system - another alternative for computer communications", Proc. Fall Joint Comput. Conf. AFIPS, 1970

[Kleinrock1975]

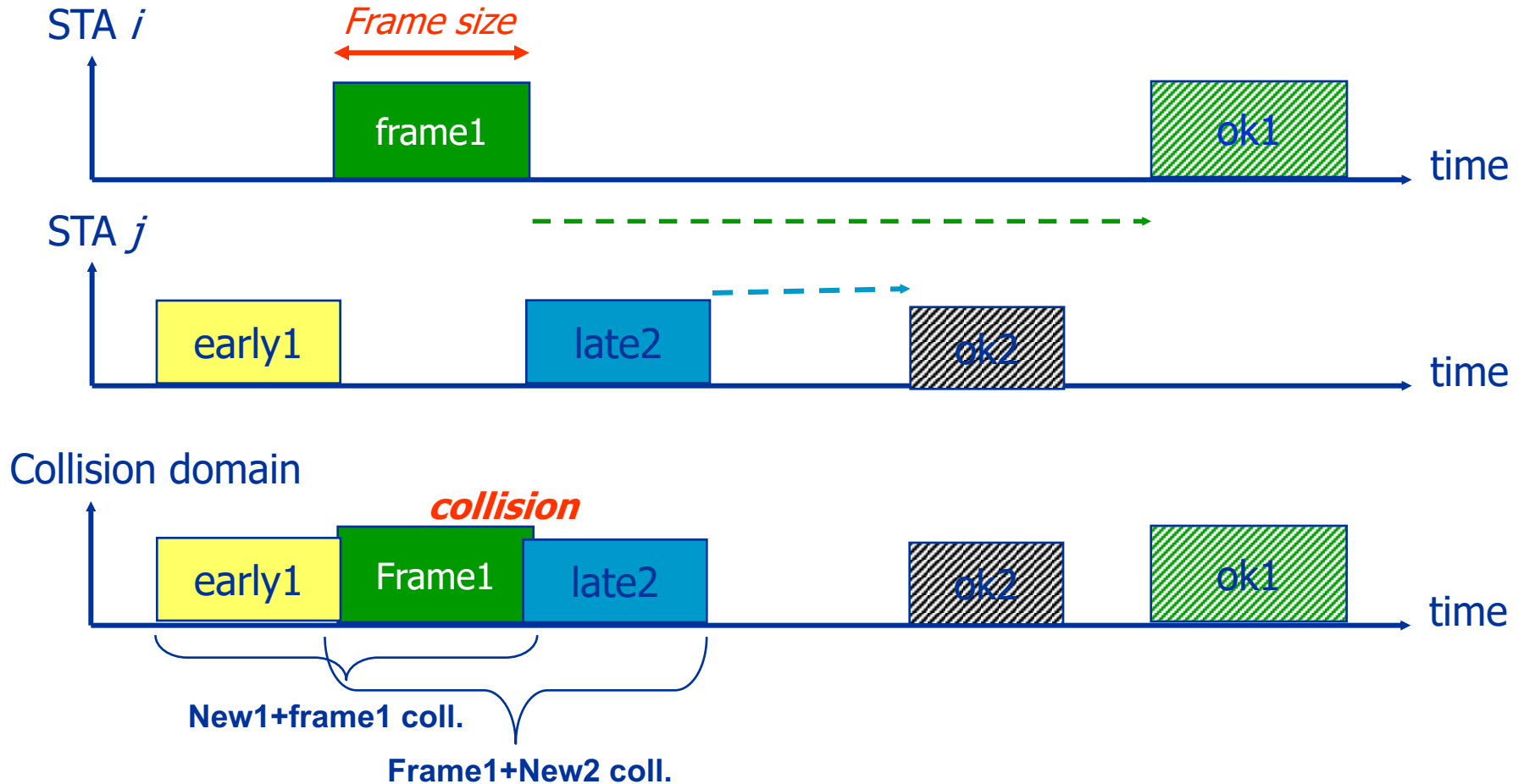
L. Kleinrock, F.A. Tobagi "Packet Switching in Radio Channels: Part I - Carrier Sense Multiple-Access modes and their throughput-delay characteristics", IEEE Transactions on Communications, Vol Com-23, No. 12, pp.1400-1416, 1975

The ALOHA protocol



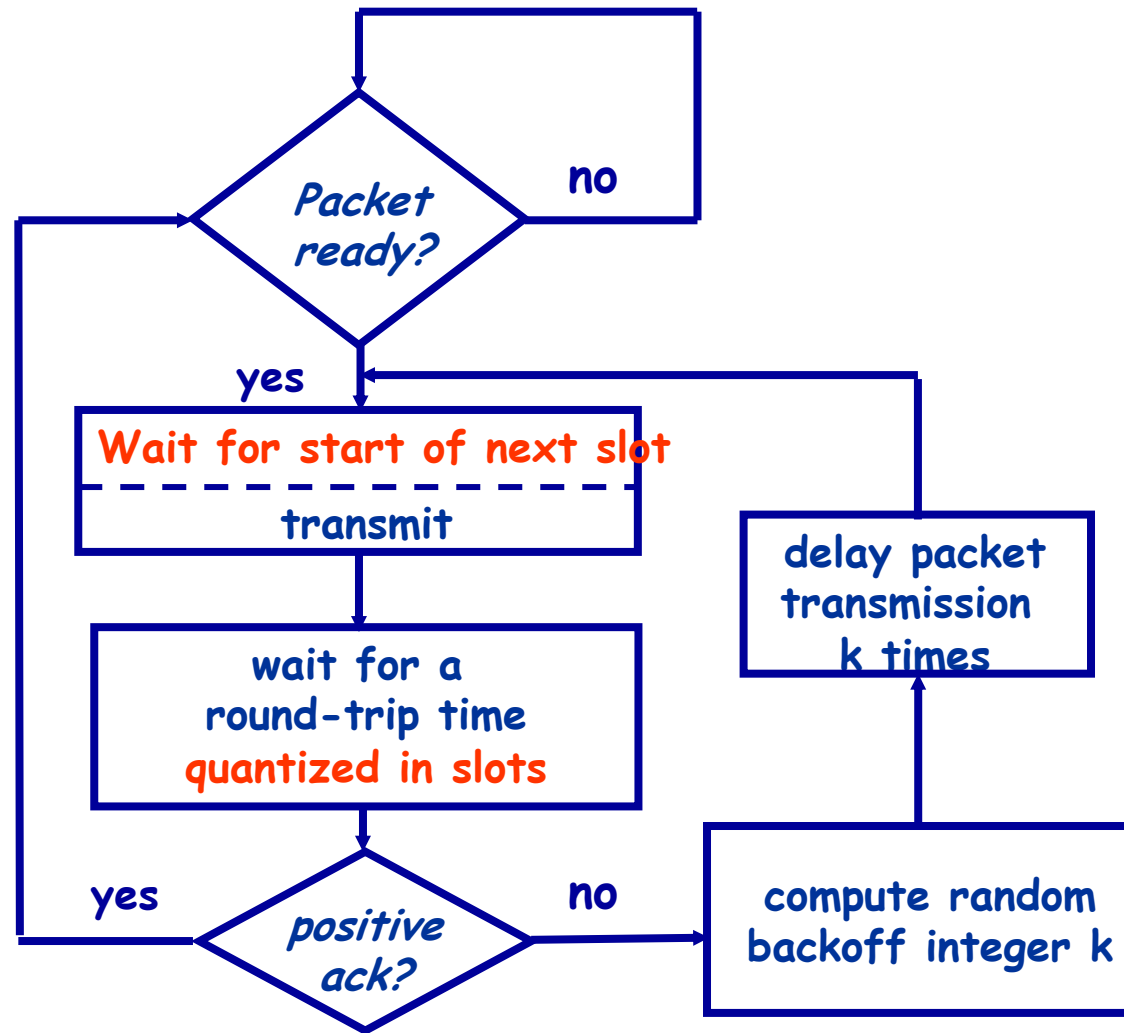
An integral part of the ALOHA protocol is feedback from the receiver
Feedback occurs *after* a packet is sent
No coordination among sources

The ALOHA protocol

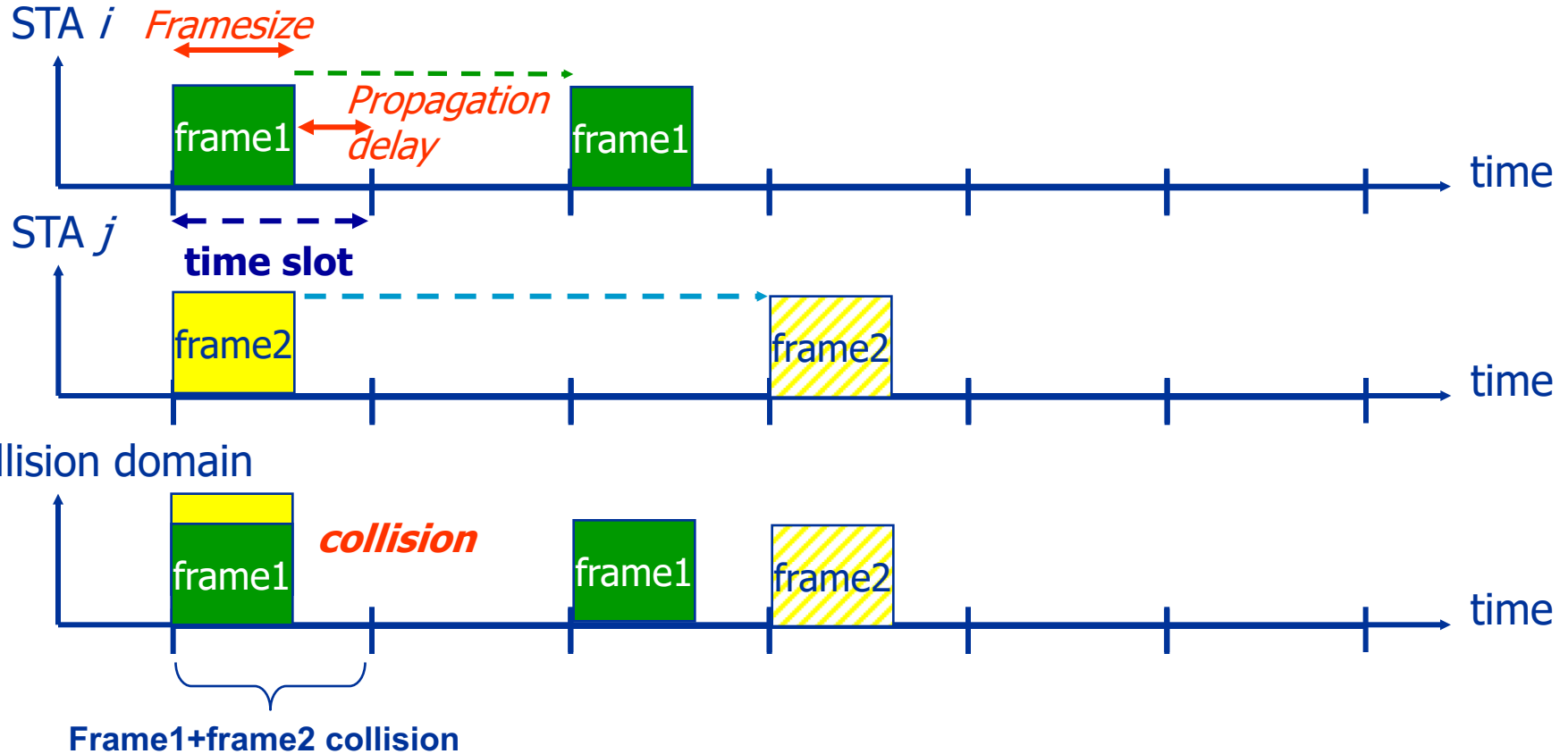


- Frame vulnerability time: twice the frame size

Slotted ALOHA

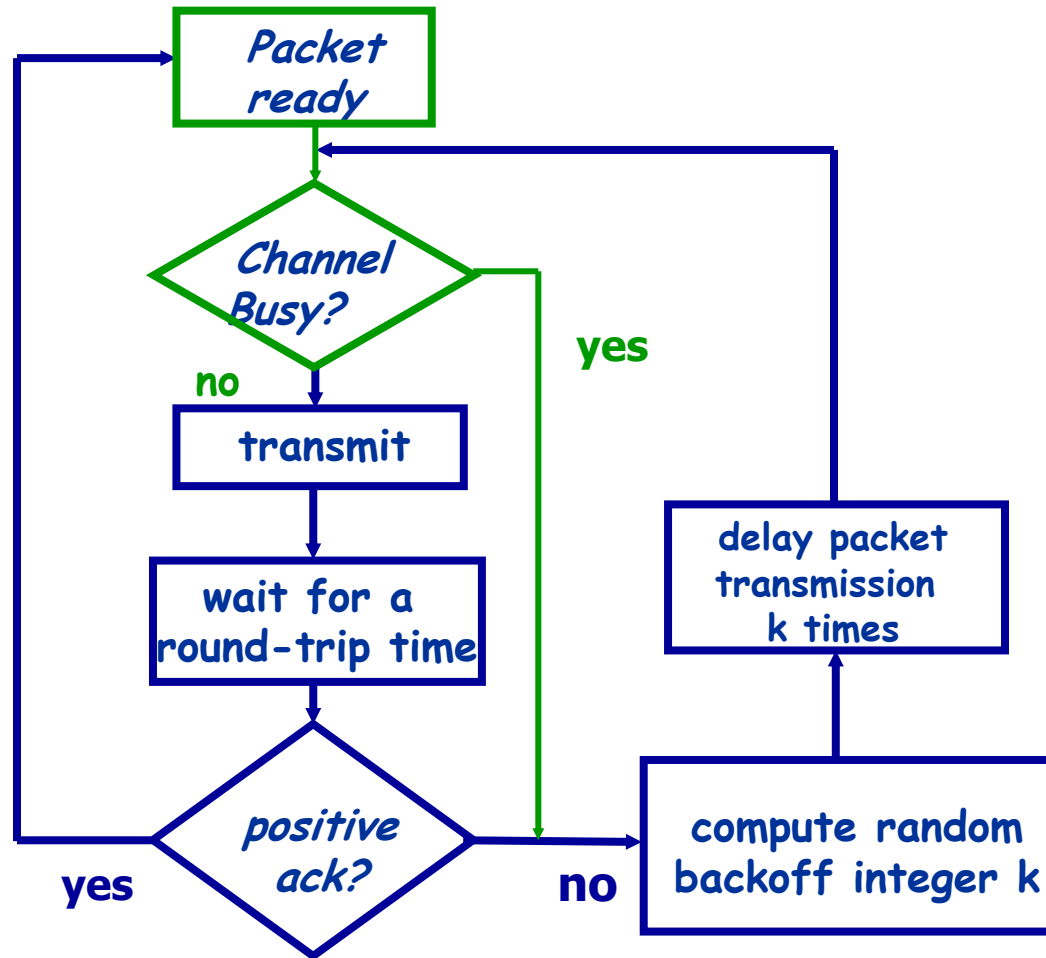


Slotted ALOHA

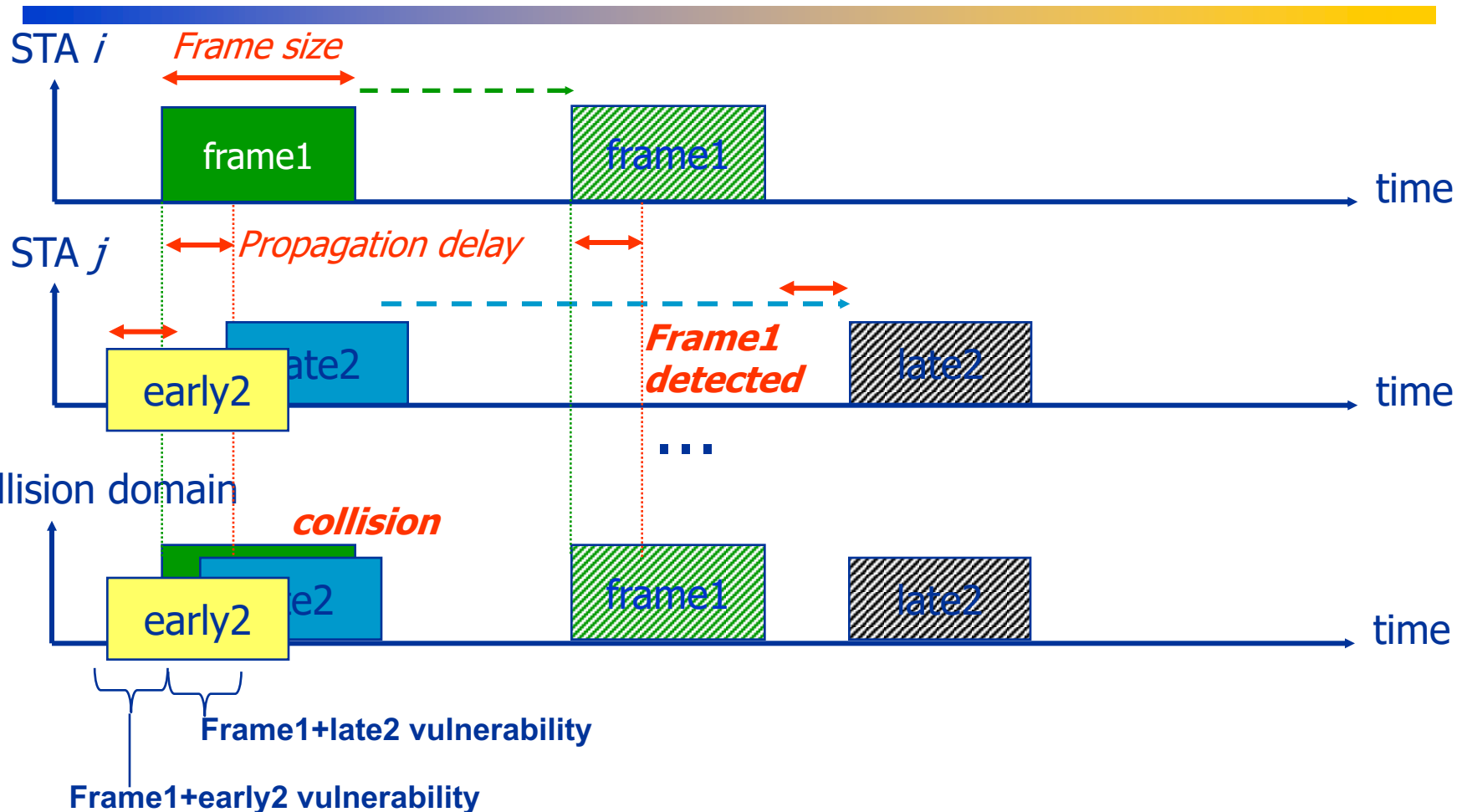


- Frame vulnerability time: the frame size (slot + propagation)

CSMA Protocol

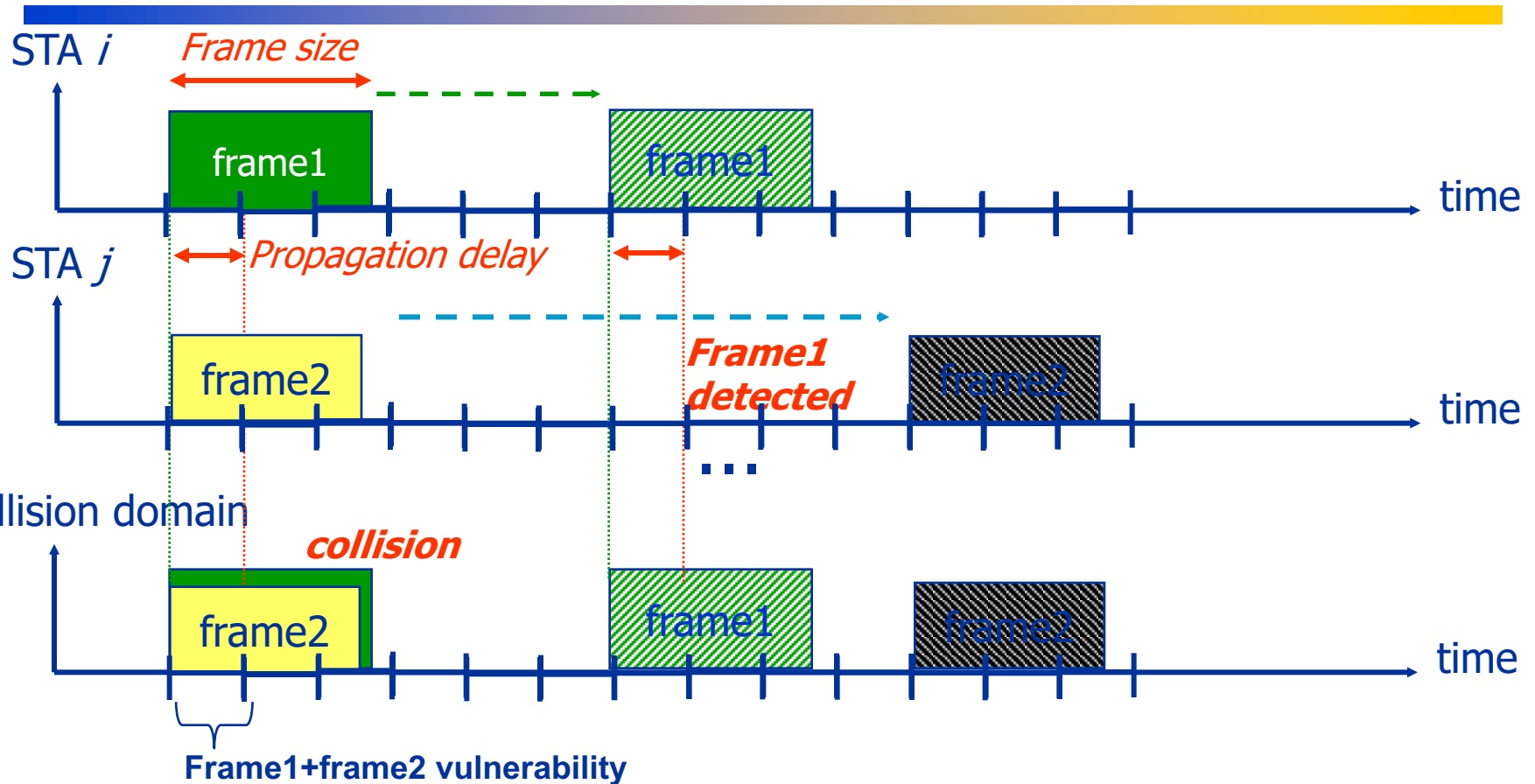


CSMA Protocol



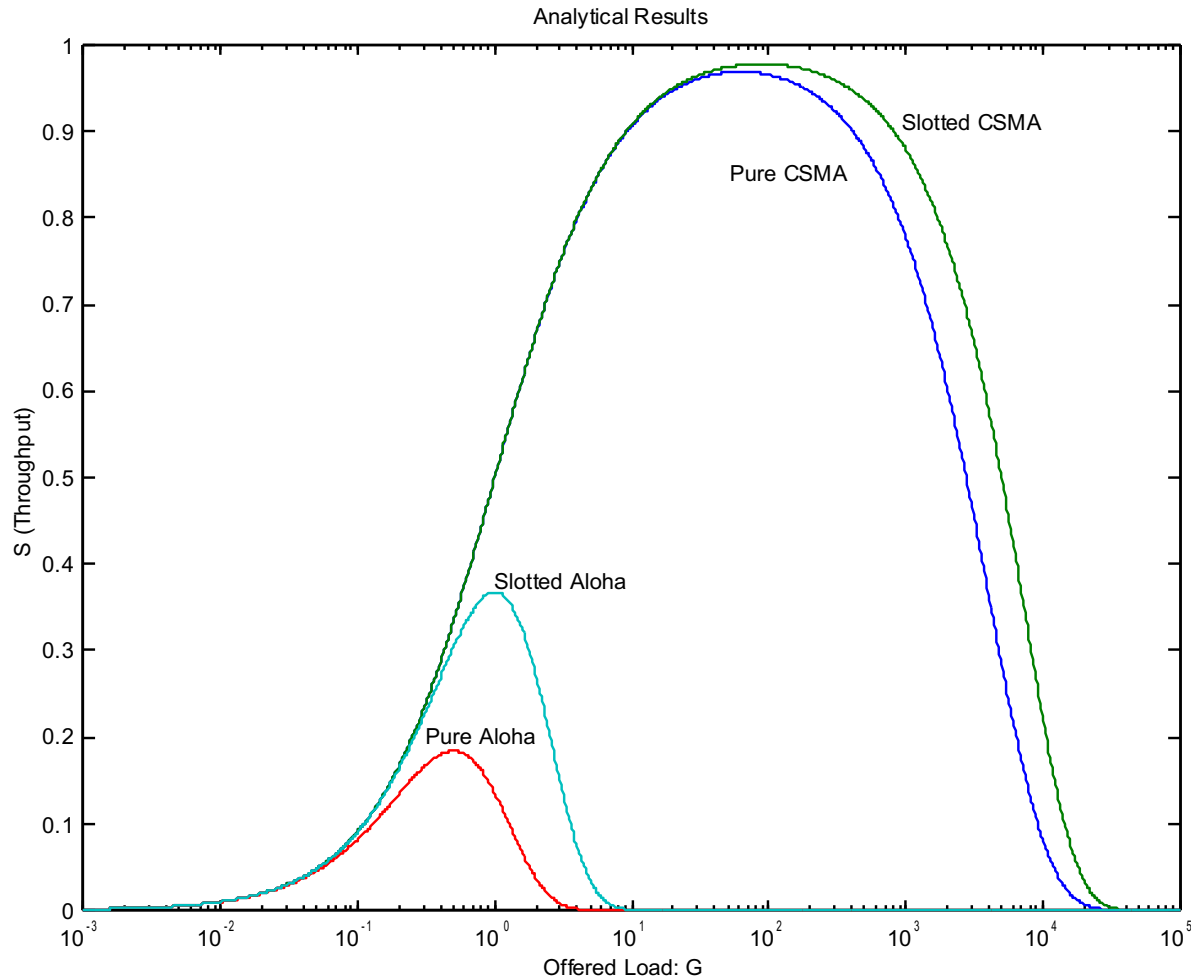
- Frame vulnerability time: twice the propagation delay

Slotted CSMA Protocol



- Frame vulnerability time: the propagation delay

Throughput comparison



Evolutionary perspective of distributed MAC

- Distributed, contention-based wireless MAC Problem:
 - the frame vulnerability (collision risk)
 - Needs resolution in distributed way (no centralized coordinator)
- let's analyze the **Space domain**
 - MACA [Karn1990]: RTS/CTS, no carrier sense (MACA-BI, RIMA...)
 - MACAW [Bharghavan et al.1994]: RTS/CTS, no carrier sense and immediate ACK (more reliable and efficient Link Layer Control)
 - FAMA [Fullmer et al.1995]: RTS/CTS, carrier sense + other stuff
- Main solution: RTS/CTS mechanism
 - Today under some criticisms

[Karn1990]

P. Karn, "MACA - A new Channel Access Method for Packet Radio", proc. 9-th Computer Networking Conference, September 1990

[Bharghavan et al. 1994]

V. Bharghavan, A. Demers, S. Shenker, and L. Zhang, "MACAW: A Media Access Protocol for Wireless LAN's," proc. ACM SIGCOMM'94, pp.212-225, London, 1994

[Fullmer et al. 1995]

C.L. Fullmer, J.J. Garcia-Luna-Aceves, "Floor Acquisition Multiple Access (FAMA) for Packet Radio Networks", Proc. ACM Sigcomm'95 Cambridge, MA, 1995

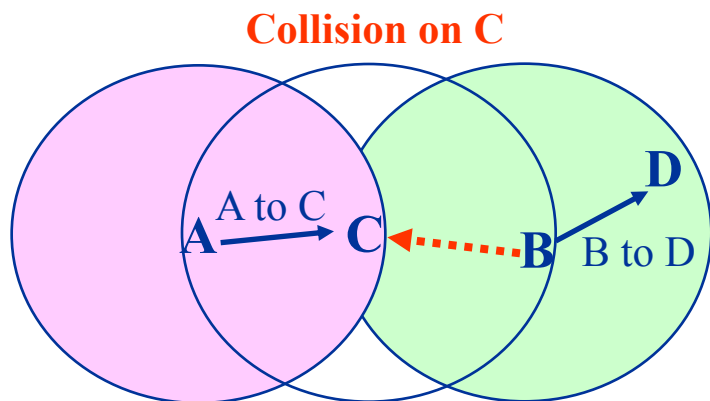
Hidden and Exposed terminals: RTS/CTS

■ The space domain:

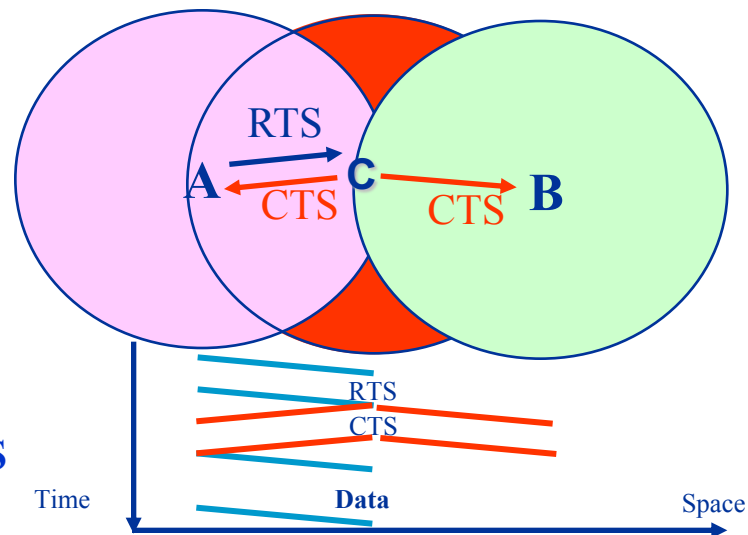
■ Hidden and exposed terminals: space vulnerability

■ RTS/CTS mechanism to contrast Hidden terminals

- Hidden terminals: B does not sense traffic, but the receiver C cannot receive its packet due to a transmission from A to C (A hidden to B)
- to seize the channel, according to CSMA/CA, a station transmits a short RTS (request to send) packet and waits for the CTS (Clear to Send) response.
 - *A transmit RTS to C and seizes the coverage area*
 - *C respond with CTS to A (B receive the CTS and does not transmit even if it cannot sense A's transmission)*

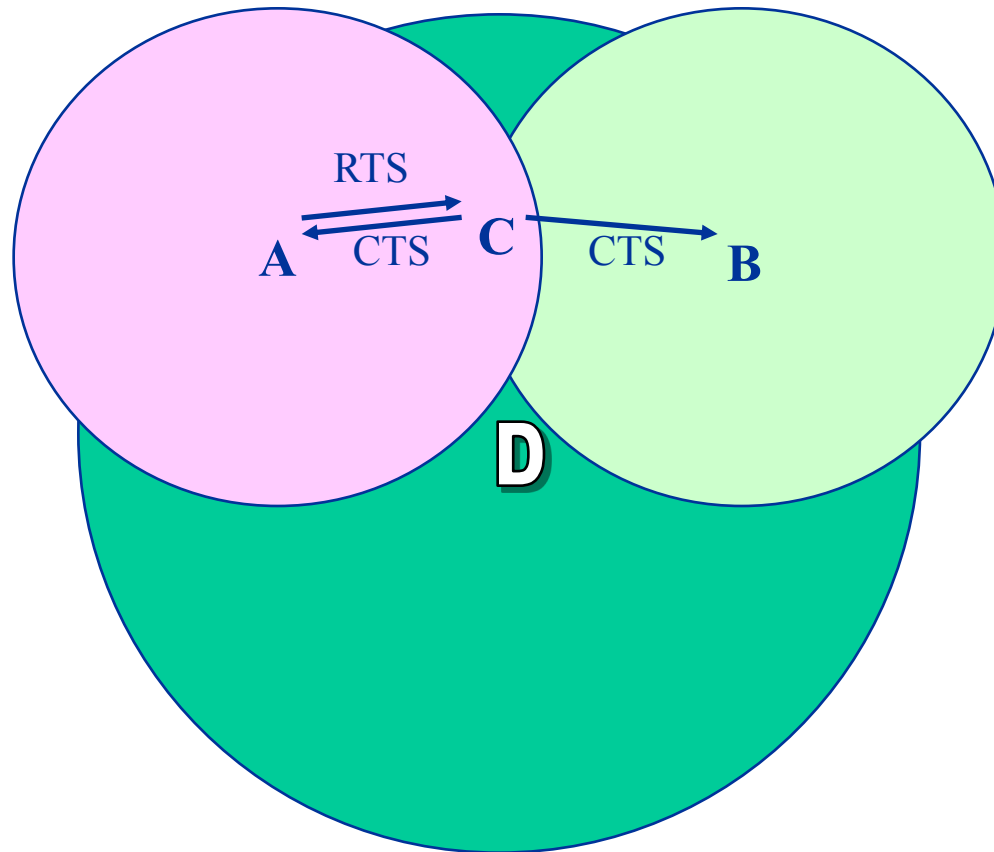


Adopt RTS/CTS



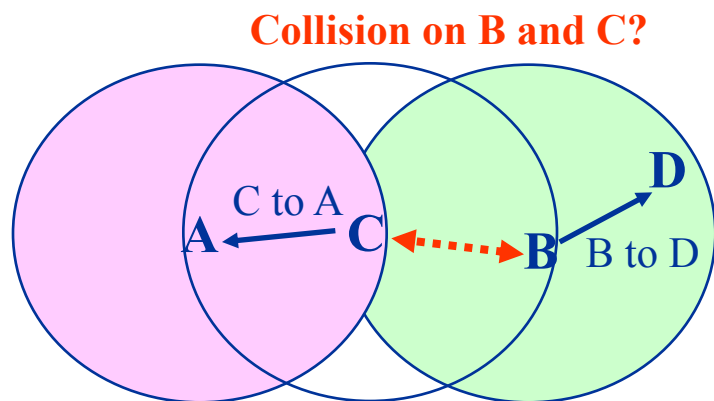
RTS/CTS drawbacks

- **RTS/CTS is not a “guaranteed” solution and it is additional overhead**
- Power asymmetry, detection and interference range $\gg$ transmission range

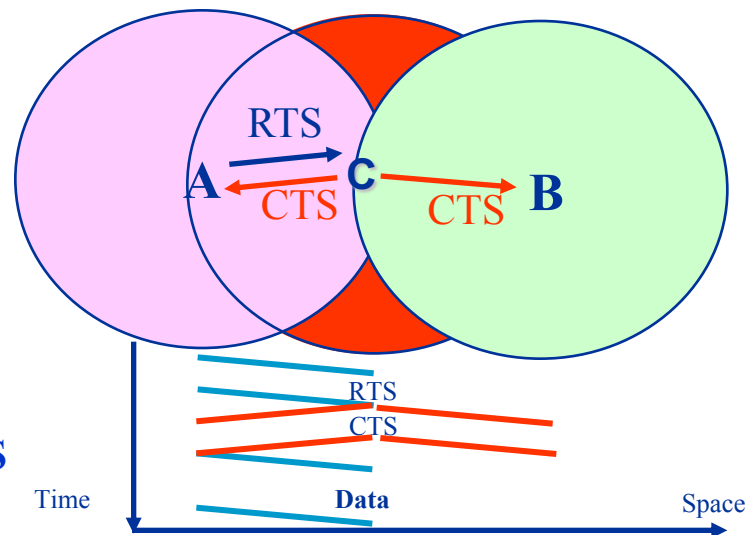


MACA: slotted RTS/CTS, no CS

- **MACA: eliminates the carrier sensing**
 - ...because the contention is on the receiver!
- Introduces **slotted RTS/CTS** (30 bytes each) and slot time equals the RTS (and CTS) duration
- Allow exploitation of concurrent spatial transmission if the receiver is not exposed to two hidden transmitter terminals
- Variations: MACA-BI, RIMA (receiver initiated)

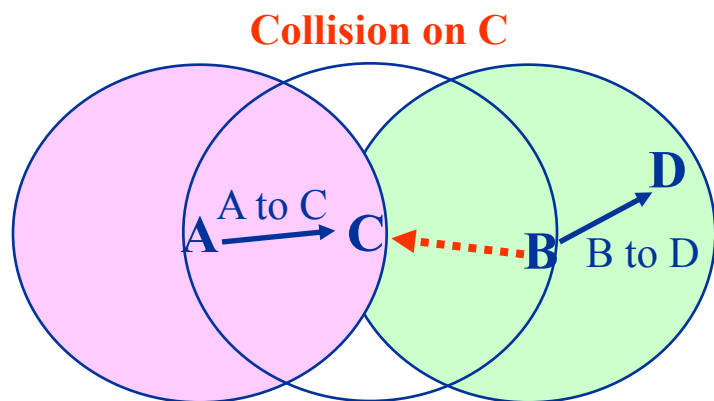


Adopt RTS/CTS

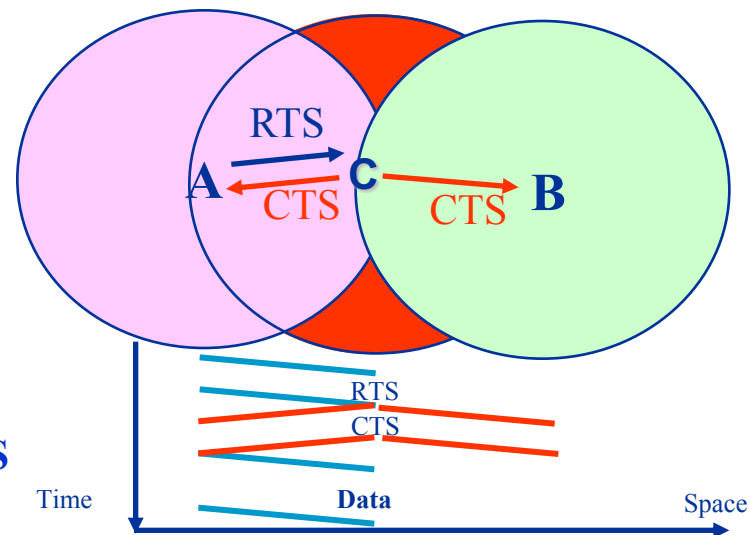


MACAW: no cs + slotted RTS/CTS + ACK

- **MACAW: fairness of the backoff procedure**
 - **MILD + Binary Exponential Backoff**
 - Cooperation-based backoff values (space-issues of contention)
- no carrier sensing before both slotted RTS/CTS
- **Introduces ACK** (RTS – CTS – DATA – ACK)
 - Efficient retransmission policy at Data Link layer
- **Problem: both sender and receiver act as receiver during frame transmissions (no concurrent space exploitation of the channel)**

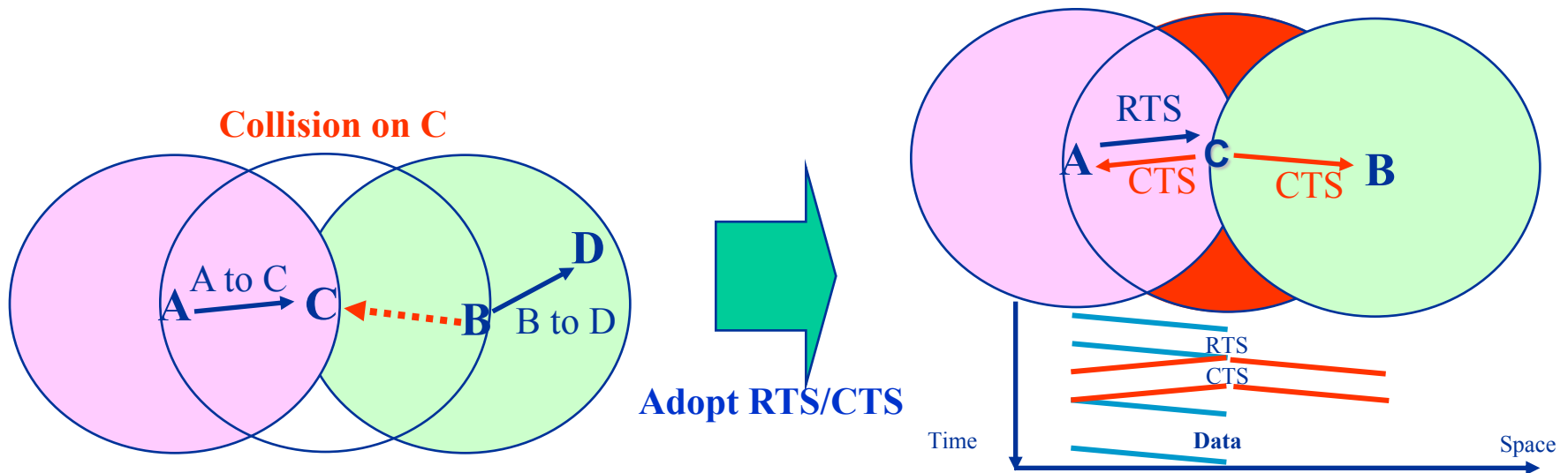


Adopt RTS/CTS



FAMA: cs + slotted RTS/CTS + ACK

- FAMA: **re-introduces carrier sensing** before both slotted RTS/CTS
- Introduces lower bound for size of RTS/CTS and CTS-dominance
- Floor acquisition: principle for time and space contention



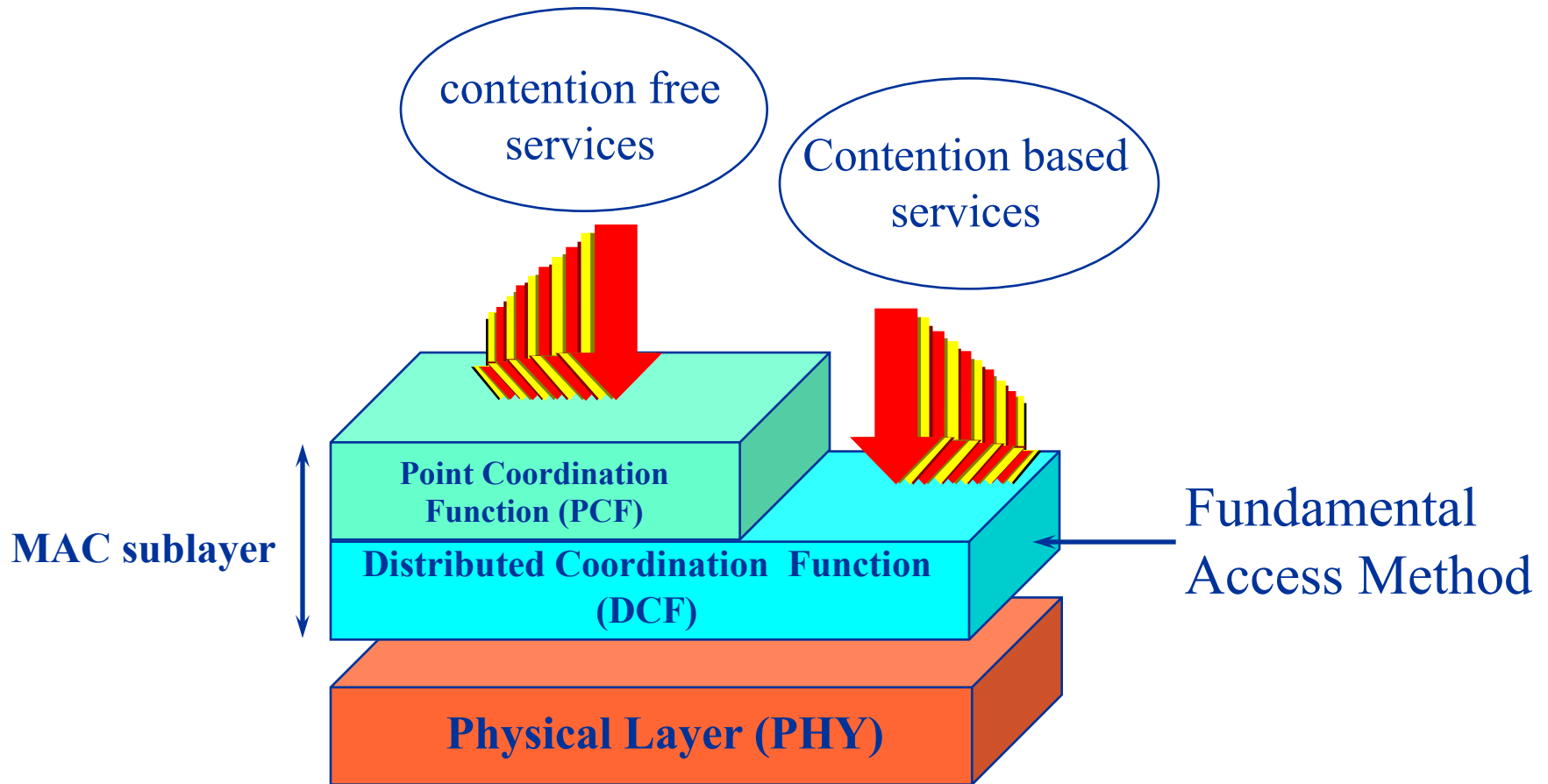
Ad hoc Multi-hop: Time/Space problems

- A bi-directional chain of MAC frames
 - TCP streams (Data + Ack)
- Self-contention (MAC layer problem)
 - Inter-stream self-contention (Data vs. Ack TCP streams)
 - Intra-stream self-contention (same TCP stream)
 - How to obtain coordination?
 - New proposed solutions
 - *Fast forward*
 - *Quick exchange*
 - *Flow numbering (pre-routing at the MAC layer???)*
 - *Frame transmission by forward invitation*

CSMA/CA: the IEEE 802.11 Wireless LAN

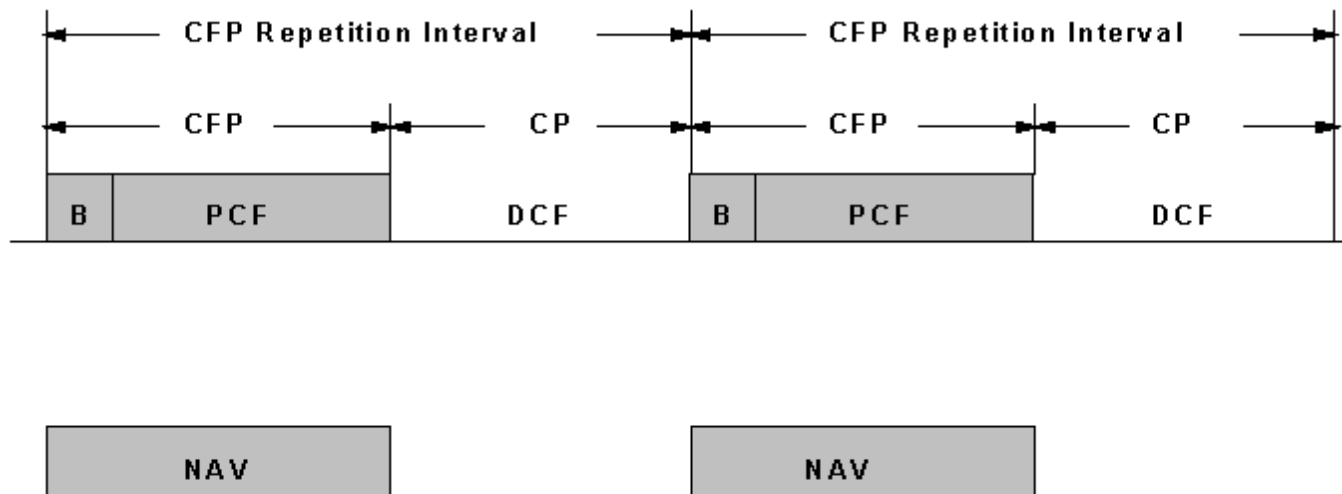
- **1 Medium Access Control (MAC) protocol:**
 - 2 coordination functions co-exist in a superframe structure (time division)
 - **Distributed Coordination Function (DCF)**
 - *Ad-Hoc networks (peer to peer)*
 - *Distributed control (no base station)*
 - *contention based access (no QoS, no minimum delay)*
 - *CSMA/CA access protocol with Binary Exponential Backoff*
 - **Point Coordination Function (PCF)**
 - *Centralized control (Base station)*
 - *Polling based access (soft QoS, minimum delay)*
 - *minimum bandwidth guarantee*

IEEE 802.11 MAC protocol architecture



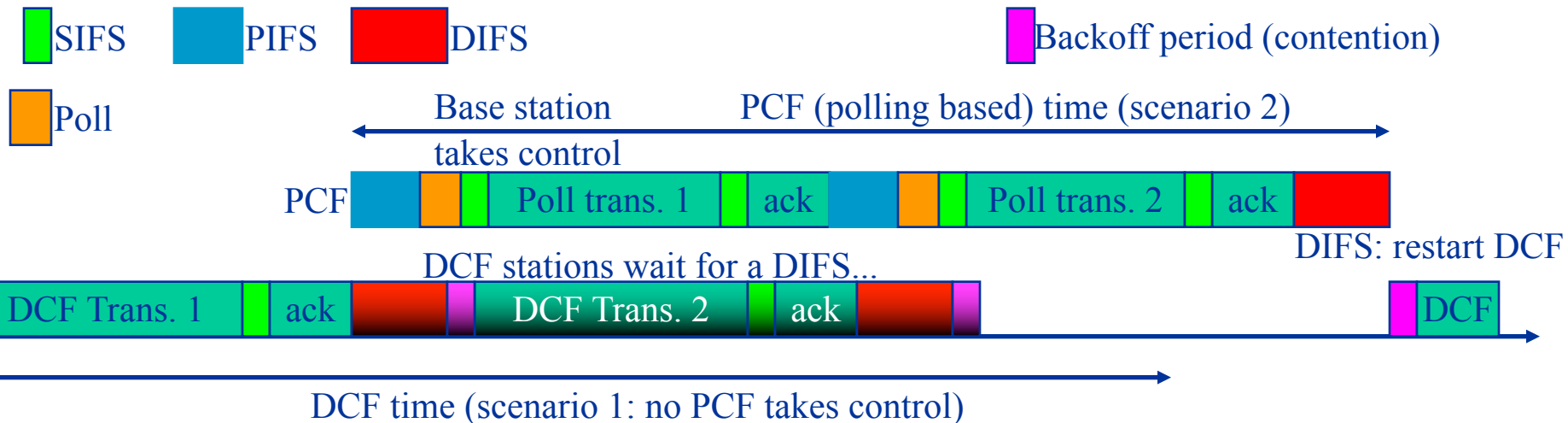
Point coordinated mode (PCF)

- point coordinated mode is a contention free, optional service
 - can co-exist with the DCF in a superframe structure.
- central coordinator, i.e. the access point (Beacon)
 - manages stations belonging to its access list.
 - guaranteed to access the channel in a contention-free environment.



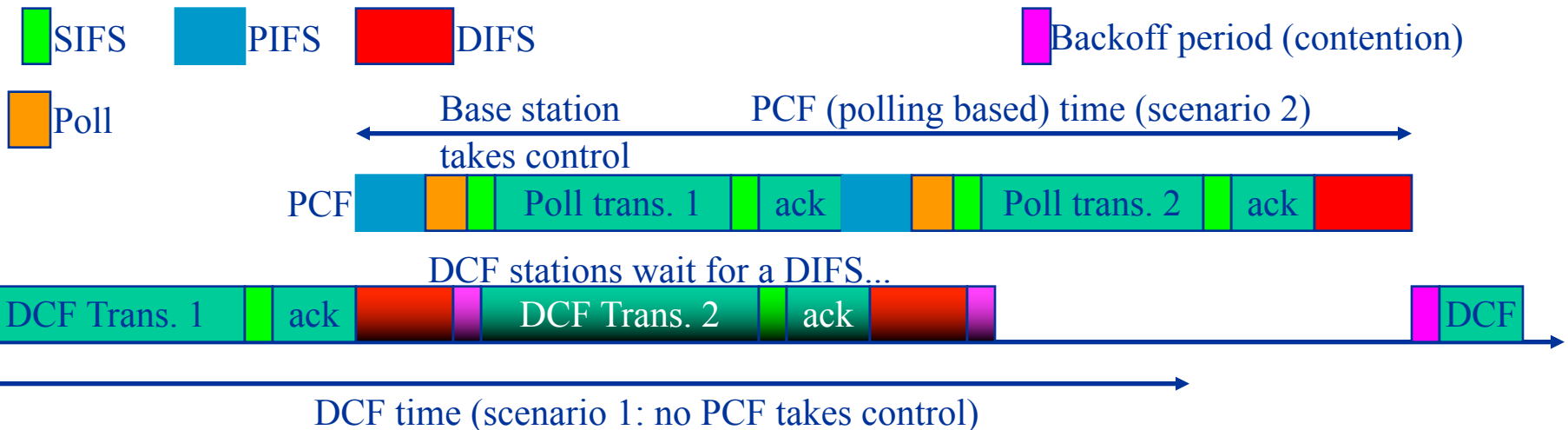
DCF and PCF control: IFS

- Each station performs a carrier sensing activity when accessing the channel
- priority is determined by Interframe spaces (IFS):
 - Short IFS (SIFS) < Point IFS (PIFS) < Distributed IFS (DIFS)
 - after a SIFS only the polled station can transmit (or ack)
 - after a PIFS only the Base Station can transmit (and PCF takes control)
 - after a DIFS every station can transmit according to basic access CSMA/CA (DCF restarts)



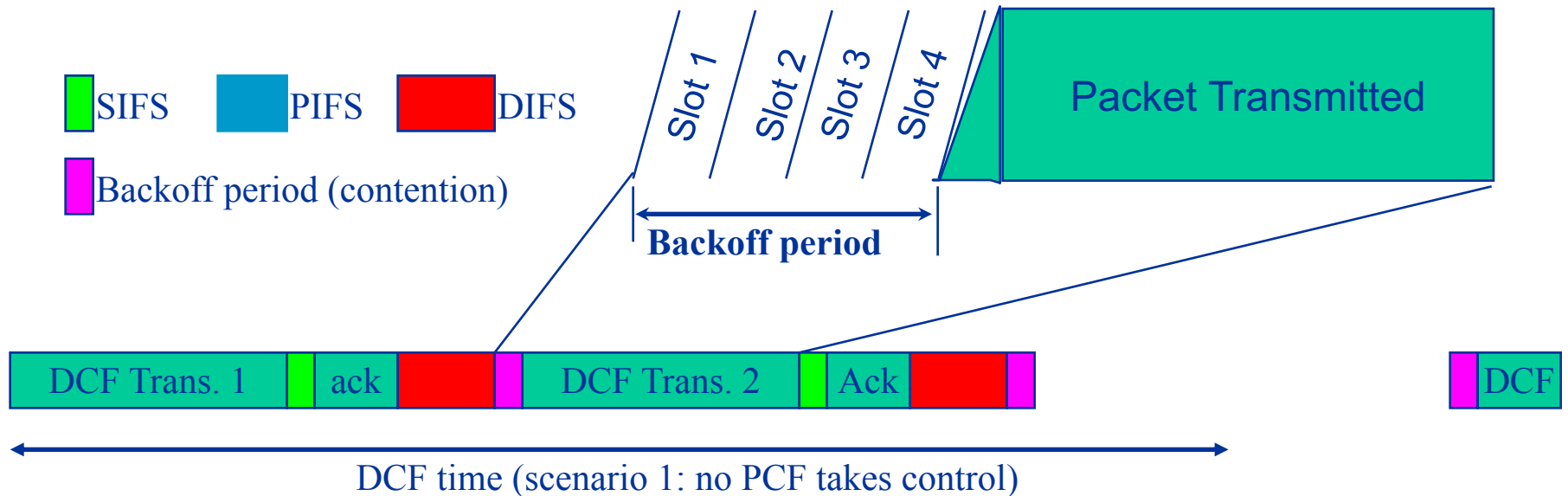
Point Coordination Function (PCF)

- during the PCF time the base station has priority in accessing the channel
 - Base Station waits for a PIFS after a transmission and takes control (DCF stations must wait for $DIFS > PIFS$)
 - base station polls stations that reserved the channel
 - at the end of the PCF period the Base Station releases the channel and DCF restarts (after a DIFS)

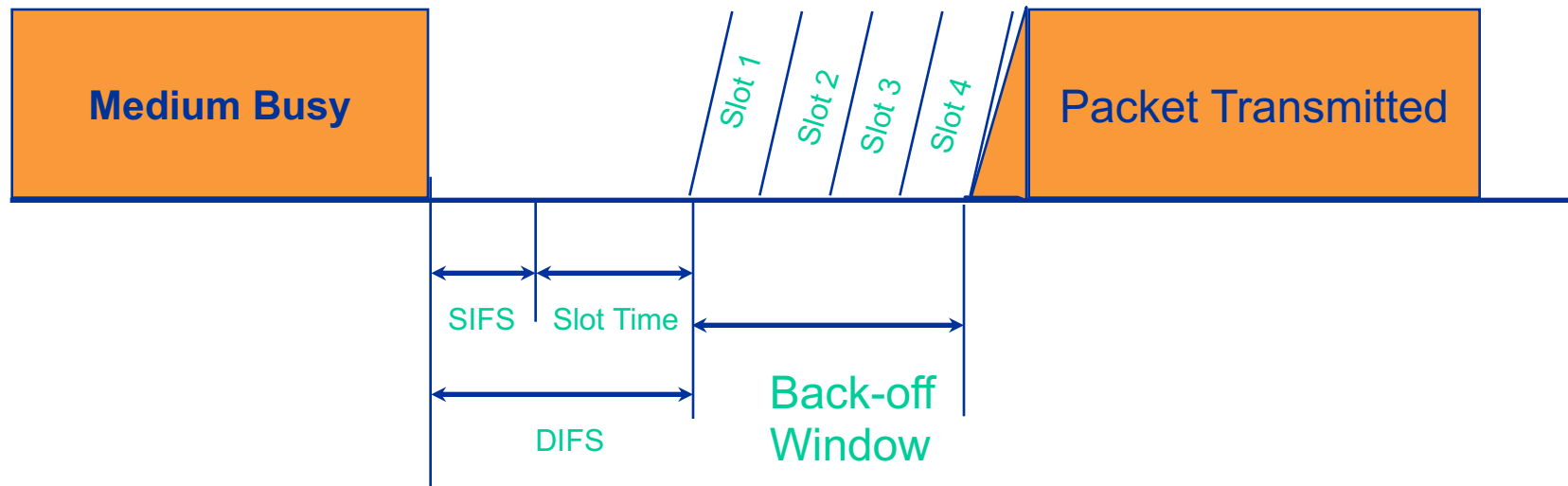


Distributed Coordination Function (DCF)

- Basic Access mode:
 - Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) access scheme (listen before transmit)
 - carrier sensing performed to detect ongoing transmissions
 - Binary Exponential Backoff over slotted idle time
 - **each station randomly selects the transmission slot in a variable sized Contention Window**
 - no Collision Detection (CD)



CSMA/CA Access Mechanism



- CSMA/CA is an efficient protocol for data traffic, like Ethernet
- Listen before transmit
- Always back-off before a transmission or retransmission
 - Designed to provide fair access to the medium

DCF Backoff procedure

- *Selection of a random Backoff Time*

CW_i = contention window size at the i -th transmission attempt. CW_i is doubled after each collision experienced (to reduce the contention)

$$\text{BackoffTime}(i) = (CW_i * \text{random}()) * \text{SlotTime}$$

i	1	2	3	4	5	6	7
CW_i	15	31	63	127	255	511	1023

- *Reduction of the Backoff Time*

After an idle DIFS period from the last transmission, a station decrements its Backoff Time by a Slot_time for each slot where **no activity is sensed on the medium**.

- *Frozen*

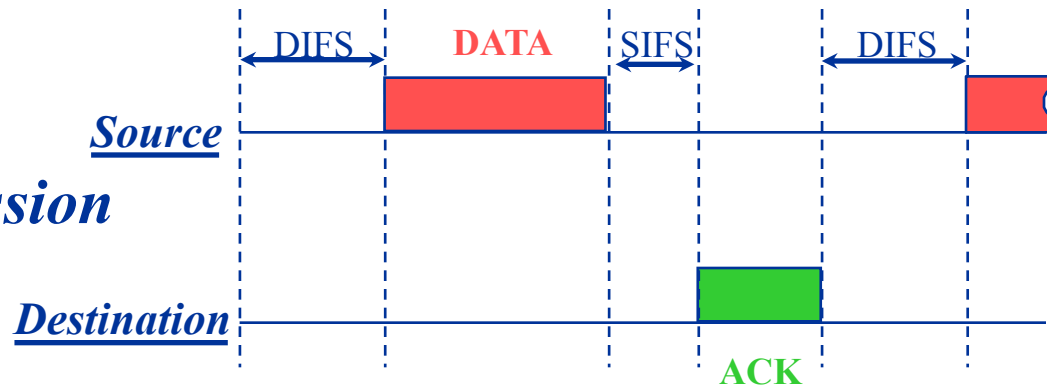
As soon as the medium is determined to be busy, the backoff procedure is suspended.

- *Transmission*

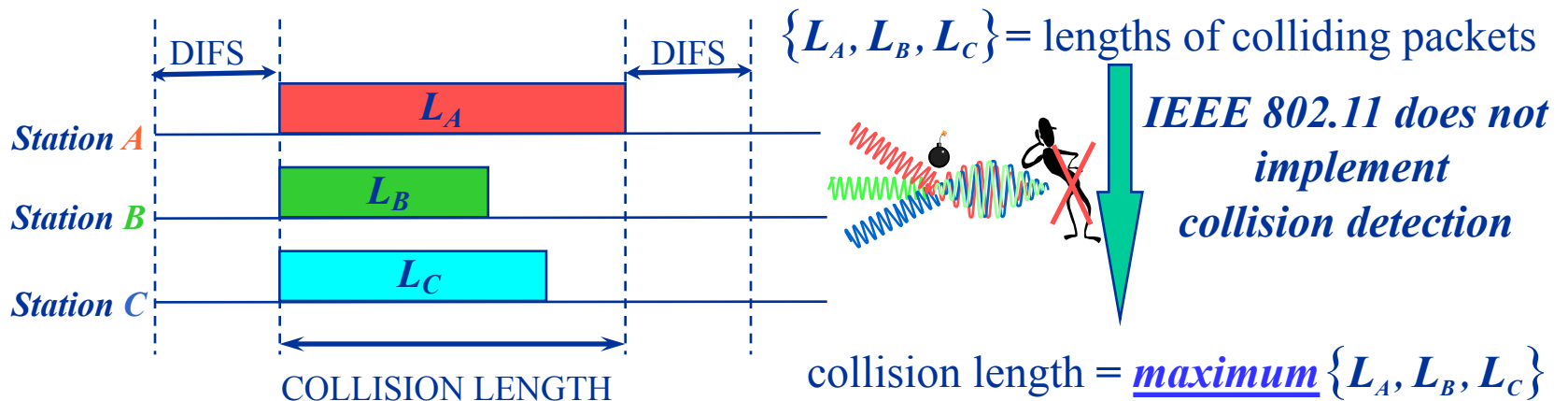
When the Backoff Time reaches zero, the station starts the transmission.

DCF basic access: overview

- *Successful transmission*



- *Collision: no CD*



IEEE 802.11 Contention Control

- Effect of high contention = many collisions

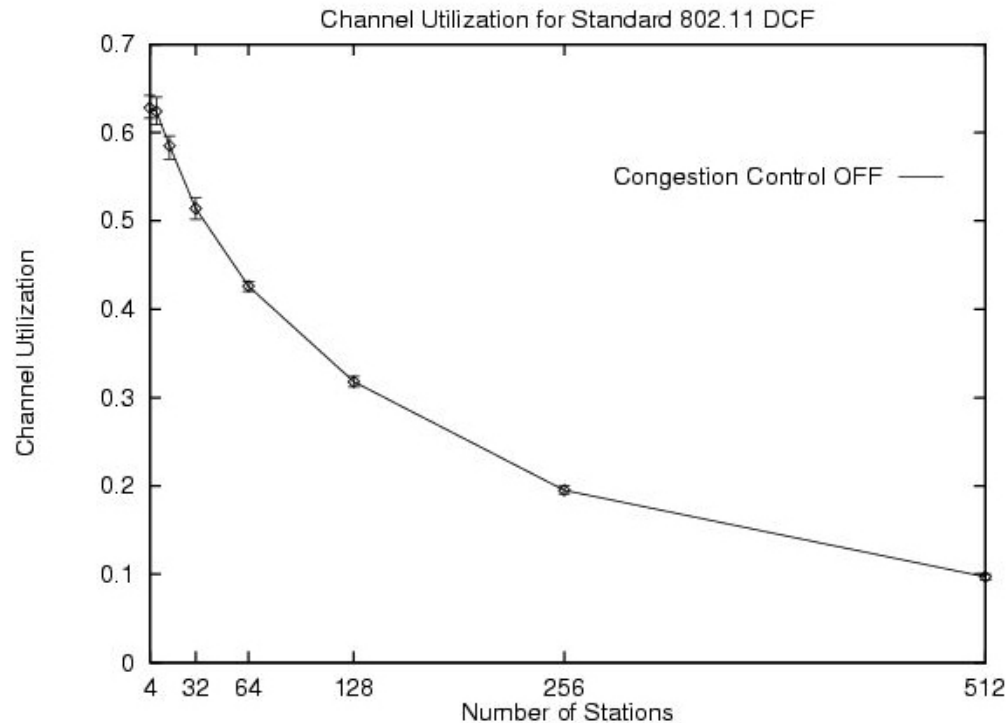


FIG. 1. Channel utilization of Standard 802.11 DCF